

HR Analytics — Employee Attendance Dashboard (AtliQ)

An end-to-end HR attendance analytics project analyzing real employee attendance data from a company called **AtliQ**, covering April–June 2022 across 74 employees. The same underlying data is analyzed five different ways — SQL, Python, Excel, Power BI, and Tableau — plus a standalone interactive web dashboard.

HR Attendance Dashboard

74 employees · April-June 2022 attendance records

EMPLOYEES

74

AVG. PRESENT/MO

14.8

AVG WFH/MO

1.8

TOTAL PAID LEAVE

160

TOTAL SICK LEAVE

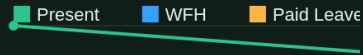
48

TOTAL UNPAID LEAVE

93.5

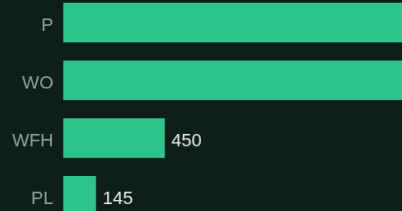
Monthly Attendance Trend

Average days per employee, per month



Attendance Status Mix

All logged days across Apr-Jun 2022



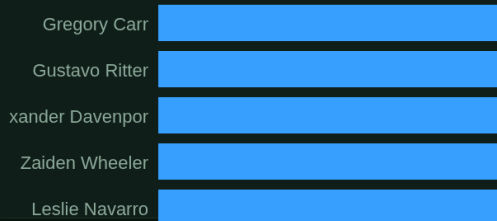
Top 10 Employees by Total Leave

Paid + sick + unpaid + bereavement + birthday + festival + menstrual leave combined



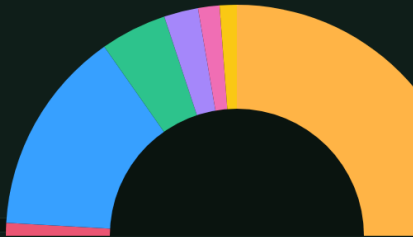
Top 10 by Work-From-Home %

WFH days as a share of all days worked



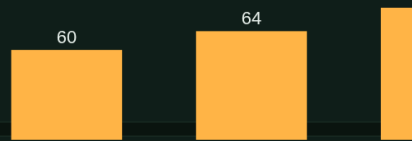
Leave Type Breakdown

Which leave categories are used most



Leave Requests by Day of Week

Excludes weekly-off/holiday days



Employee Detail

All 74 employees — sorted by total leave, highest first

EMPLOYEE CODE	NAME	TOTAL PRESENT	TOTAL WFH	TOTAL LEAVE
Atq -419	Kaylah Schultz	29	10	17
Atq -372	Miley Combs	47	0	16
Atq -452	Ayanna Atkins	40	3	14
Atq -349	Trystan Ortega	44	0	13
Atq -404	Ana Little	43	0	12
Atq -335	Madelynn Gray	33.5	4	11.5
Atq -485	Gregory Carr	0	11	10
Atq -396	Joel Cruz	27	22	9.5
Atq -393	Emma Freeman	43	4	9
Atq -398	Bo Cordova	37	18	9

Data note: 24 employee codes in the source workbook are associated with more than one name across different months (e.g. Atq-335 appears as both "Tori Shannon" and "Madelynn Gray"). This looks like a data-entry inconsistency in the original attendance sheet rather than a real employee change, and is preserved as-is here rather than silently merged — worth a data-cleaning pass before this feeds a production HR system.

Project description

HR and people-ops teams need visibility into attendance and leave patterns to plan workforce capacity, spot burnout risk, and catch disengagement early. This project answers:

- Who is present, on leave, or working from home, and how does that shift month to month?
- Which employees are taking the most leave, and of what type?
- Are there day-of-week patterns in leave requests?
- Is attendance trending up or down over the quarter?

Headline numbers:

Metric	Value
Employees tracked	74
Months covered	April, May, June 2022
Avg days present / employee / month	14.8
Avg WFH days / employee / month	1.8
Total paid leave logged	160 days
Total sick leave logged	48 days
Total unpaid leave (LWP) logged	93.5 days

Technology used

Layer	Tool
Data cleaning & consolidation	Python (pandas)
Exploratory analysis	Python (pandas, matplotlib, seaborn)
Querying	SQL (SQLite-tested, ANSI-compatible)
Reporting workbook	Excel (openpyxl, live formulas)
BI dashboard	Power BI (Power Query M, DAX)
BI dashboard (alt)	Tableau (calculated fields)
Web dashboard	HTML + Chart.js

Project structure

```
hr-analytics-attendance/  
├─ data/           source workbook + cleaned CSVs  
├─ sql/            SQL analysis queries  
├─ python/         data cleaning + EDA scripts  
├─ excel/          KPI workbook with live formulas
```

— powerbi/	real .pbix report + DAX reference
— tableau/	calculated fields + build guide
— dashboard/	standalone interactive HTML dashboard
— images/	charts exported from Python EDA
— screenshots/	dashboard preview image
— README.md	

How to use this project

View the dashboard instantly (no setup) Download `dashboard/hr_dashboard.html` and open it in any browser — no installation required.

Run the Python analysis

```
git clone https://github.com/aastha1534/hr-analytics-attendance.git
cd hr-analytics-attendance
pip install -r python/requirements.txt
python python/consolidate_attendance.py
python python/eda_attendance.py
```

Query the data with SQL Load `data/attendance_monthly.csv` and `data/attendance_daily.csv` into SQLite (or any SQL database) and run the queries in `sql/hr_attendance_queries.sql`.

Open the Excel workbook Open `excel/HR_Attendance_Analysis.xlsx` directly — the Dashboard sheet recalculates automatically from the RawData sheet.

Open the Power BI report Open `powerbi/HR-Analytics-Atliq.pbix` in Power BI Desktop — it's a fully built, working report.

Rebuild in Tableau Follow `tableau/README.md` — includes every calculated field and a sheet-by-sheet build guide.

Key insights

- **Attendance dropped over the quarter:** average days present per employee fell from 17.8 (April) to 16.4 (May) to 10.4 (June).
- **WFH held steady** at roughly 1.7–2.0 days/employee/month across all three months.
- **Leave is concentrated in a small group** — the top 10 leave-takers account for a disproportionate share of total leave days.
- **A handful of employees work from home almost exclusively.**
- **Leave requests cluster around certain weekdays** — a pattern worth watching for approval-policy implications.

Data quality note

The source workbook links **24 employee codes to more than one name** across different months (e.g. `Atq-335` appears as both "Tori Shannon" and "Madelynn Gray"). This is preserved as-is throughout the

project rather than silently merged, since it looks like a data-entry inconsistency worth a cleaning pass before this feeds a production HR system.

About the data

This project uses the real AtliQ attendance workbook as provided — daily attendance codes (Present, WFH, Paid/Sick/Unpaid Leave, Weekly Off, Holiday, and specialty leave types) for 74 employees across April–June 2022, plus the original Power BI report already built on it.